Compensating maize price shocks under imperfect information: evidence from Malawi IHS6

Technical research report: data, survey design, welfare model, targeting rules, statistical methods, validation and results

PolicyPath project · October 2026 · Validated run `maize_20261025T173130_1dafdd` (200 bootstrap replicates; 10 synthetic-fixture tests, 80 expectations passed; feasibility gate passed) · All figures are simulations under stated scenarios, not causal estimates.

Abstract. Using the Malawi Sixth Integrated Household Survey (IHS6, 2024/25), we simulate the first-order distributional effects of a maize price increase. The model allows separate retail and farm-gate price changes. We then compare compensation rules at an identical fiscal budget under three objectives: the poverty gap, the squared poverty gap, and the compensated share of initially poor households' losses. Official IHS5 and IHS6 poverty statistics are first reproduced exactly from the microdata. In a central scenario (retail grain +25%), losses fall from 4.1% of consumption in the poorest decile to roughly zero in the richest, and the poverty gap rises from 15.72 to 16.69. Among rules a government could design, a same-round cross-fitted proxy means test (PMT) achieves about 75% of the perfect-information reduction in the poverty gap, against 48% for an equal per-person transfer and 54% for district allocation. Adding maize-exposure prediction to the PMT has no measurable effect: the paired difference is within ±0.003 on all objectives, with 95% CIs spanning zero. District membership explains only 6–11% of household variation in losses and post-shock poverty. The two perfect-information benchmarks tie on the poverty gap but diverge sharply on the squared gap and on loss compensation, so the choice of objective is decisive. PMT results are upper bounds for a deployable registry.

1. Data

Dataset	Reference	Fieldwork	Households	EAs	Use
IHS5	MWI_2019_IHS-V_v06_M	Apr 2019 – Apr 2020	11,434	717	Stage 1 replication
IHS6	MWI_2024-2025_IHS-VI_v01_M	Jul 2024 – Jul 2025	12,502*	782*	Replication and all Stage 2A analysis

* The national consumption aggregate excludes the 386 Dzaleka refugee-camp households (24 EAs), as in official statistics. Source: National Statistical Office via the World Bank Microdata Library. The microdata are licensed for use within the user's organisation and may not be redistributed, so they are not included in the reproducibility package.

Modules used:

- the consumption aggregate (real annual consumption, poverty lines and flags, deflators);
- the household cover (strata, camp flag);
- food consumption G1 (7-day recall: item, purchased quantity, amount spent);
- crop sales: rainy season I and dimba O (value and month or year of sale);
- crop cultivation G and M;
- plot areas C;
- household roster B, education C, dwelling F and assets L (PMT covariates);
- reference season X.

2. Stage 1: replication of official poverty statistics

Using the released aggregates and person weights (household weight × household size), every published IHS6 figure was reproduced to the published decimal: 103 of 103, covering the national, urban/rural, regional, rural-regional, sex-of-head and all 32 district headcount and ultra-poverty rates, plus the gaps, Gini indices and poverty lines.

For IHS5, 19 benchmarks were found and all reproduce once the regional figures are read as whole-region rates. *Malawi in Figures 2023* labels them "rural" but they match whole regions exactly, and NSO's 2026 slides label them "Region". The published IHS5 ultra-poverty rate of 20.8% conflicts with NSO's own 20.5%; the data give 20.45.

Design-based standard errors.

- PSU = enumeration area (EA).
- Strata: the released IHS6 `strata` variable (= district). IHS5 strata were reconstructed as district.
- National poverty SE: 0.81 (IHS5) and 0.89 (IHS6) percentage points, with a design effect of about 1.8–2.0.

3. Welfare model

Household *h* buys maize products *k* at retail prices and sells grain at farm-gate prices. For small price changes, with quantities fixed (envelope theorem; Deaton 1989), the money-metric welfare change in real July 2024 MWK per year is:

$$\Delta y_h = \lambda\pi \cdot S_h - \pi \cdot B_h^{grain}(1 + \tfrac{1}{2}\varepsilon\pi) - \theta\pi \cdot B_h^{flour}(1 + \tfrac{1}{2}\varepsilon\theta\pi)$$

where:

- *S* is the real value of maize sales (rainy plus dimba season);
- *B* is the annualised (×52) real value of 7-day maize purchases (grain; mgaiwa, refined and madeya flours);
- π is the retail grain price change, θ the flour pass-through, λ the farm-gate transmission and ε the compensated own-price elasticity.

Own consumption drops out. With a price band, a household that is autarkic at the margin is unaffected at first order, so the model uses transaction values only. No quantity conversion factors are needed; IHS6 doesn't release them.

Deflation. Purchases are deflated with each household's combined spatial × temporal price index (verified: index = spatial × temporal to 7e-6). Sales are deflated with the area × month temporal index at the month of sale. Sales made before the fieldwork window are assigned the window-start index.

Post-shock per-capita welfare is $(c_0 + \Delta y)/n$, compared with the national line (1,562.57 MWK per person per day) and the food line (1,071.11).

Central scenario: π = 0.25, λ = 0.5, θ = 0.85, ε = 0; green maize excluded.

3.1 Feasibility gate

All essential checks passed before any policy model ran:

- poverty flags recomputed with 0 mismatches;
- deflator identities hold;
- maize purchase value missing in 25 of 8,912 purchase rows;
- annualised purchases never exceed nominal food consumption;
- sale values complete and all sales dated;
- covariate missingness ≤ 0.01%.

Plausibility: 74.6% of households grow maize, against IFPRI's 74% for IHS5.

4. Compensation rules and objectives

Budget. Each scenario has a budget G = κ × the aggregate loss of losing households (central κ = 0.5, i.e. 1.127×10^{11} MWK). Every rule spends exactly G under the weights used for evaluation; this is asserted to 1e-8 in every scenario and bootstrap replicate.

Rule	Allocation	Information class
R1 universal	Equal per person	Implementable: population counts
R2 geographic	District share ∝ survey-estimated post-shock gap mass; equal per person within district	Implementable: district survey estimates
R3 PMT	Equal per person to the lowest predicted consumption, up to coverage q	Survey-ideal (same-round cross-fit)
R4 exposure	The same, ranked by most negative predicted per-capita Δy	Survey-ideal
R5 PMT + exposure	The same, ranked by Duan-smeared predicted consumption + predicted Δy	Survey-ideal
B1	*Level up the poorest to a common floor ≤ z*	*Perfect information (infeasible)*
B2	*Reimburse losses of post-shock poor*	*Perfect information (infeasible)*
B3	*Reimburse losses of initially poor*	*Perfect information (infeasible)*

Coverage. q equals the government's survey estimate of the post-shock headcount.

Prediction models. Weighted least squares (person weights) on:

- household size and composition, and head's sex, age and qualification;
- wall, roof and floor materials, rooms per person and electric or solar lighting;
- ten assets, land per person and maize cultivation;
- district and urban fixed effects.

The models are fitted with **EA-grouped 5-fold cross-fitting**: no household is scored by coefficients estimated on its own EA.

Objectives (person-weighted, after shock and transfer):

- P_1 = poverty gap × 100 (primary);
- P_2 = FGT2 × 100;
- loss compensation = Σ_{poor_0} w·min(T, loss) / Σ_{poor_0} w·loss.

Attainable share = (no-transfer value − rule value) / (no-transfer value − benchmark value). The benchmark is B1 for P_1 and P_2, and B3 for loss compensation.

5. Statistical methods and uncertainty

Bootstrap. Rao–Wu rescaled bootstrap (`survey::as.svrepdesign`, type "subbootstrap", 200 replicates), resampling n_h − 1 EAs within each released stratum. The treasury budget is fixed in MWK across replicates. Components:

- **U3 (primary, joint):** the government re-estimates its PMT and exposure models, coverage and district estimates from each replicate, and outcomes are evaluated with the same replicate weights.
- **U1:** inputs fixed, replicate evaluation (sampling only).
- **U2:** inputs re-estimated, full-sample evaluation (information error only).

Paired differences between rules are computed within replicates. Percentile 95% intervals are reported.

Deterministic sensitivity grid (972 scenarios):

Parameter	Values
π	0.10, 0.25, 0.50
λ	0, 0.5, 1
θ	0.7, 0.85, 1
ε	0, −0.3, −0.6
κ	0.25, 0.5, 1
Green maize	excluded / included
Purchase annualisation	×52 / seasonally normalised (area × calendar quarter)

District diagnostics. Design-based district means with Taylor-linearised SEs, bootstrap rank probabilities, and between-district R^2 of household outcomes.

6. Results

6.1 Incidence (central scenario)

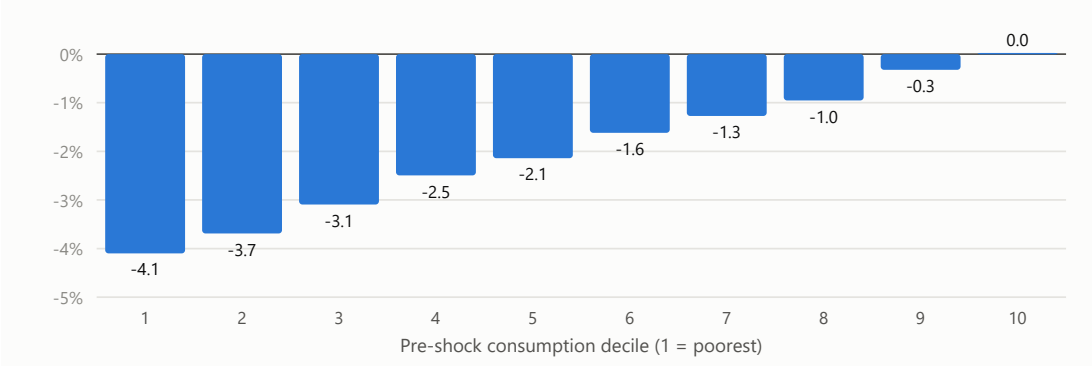


Figure 1. $\Delta y/c_0$ (%) by pre-shock consumption decile.

Shock without compensation	Pre	Post
Headcount P_0 (%)	47.26	48.68
Poverty gap P_1 (×100)	15.72	16.69
Squared gap P_2 (×100)	7.13	7.76
Ultra-poverty (%)	24.04	26.00

50.1% of people are in losing households and 14.1% in gaining ones. The aggregate net change is −1.04% of consumption. Net buyers are 67% of the poorest decile and 23% of the richest.

6.2 Compensation rules (central scenario, κ = 0.5)

Rule	P_1	P_2	Loss comp.	Share P_1	Share P_2	Share loss	Exclusion error	Inclusion error
R0 none	16.69	7.76	0.000	0	0	0	1.00	–
R1 universal	16.23	7.44	0.288	0.48	0.26	0.29	0.00	0.51
R2 geographic	16.17	7.39	0.328	0.54	0.31	0.33	0.00	0.51
R3 PMT	15.98	7.23	0.438	0.75	0.45	0.44	0.24	0.24
R4 exposure	16.11	7.34	0.374	0.61	0.35	0.38	0.38	0.38
R5 PMT + exposure	15.97	7.23	0.439	0.75	0.45	0.44	0.24	0.24
B1	*15.74*	*6.57*	*0.141*	*1*	*1*	*0.14*	–	–
B2	*15.77*	*7.17*	*0.918*	*0.97*	*0.50*	*0.93*	–	–
B3	*15.74*	*7.12*	*0.991*	*1.00*	*0.53*	*1*	–	–

Exclusion error = share of post-shock poor persons receiving nothing. Inclusion error = share of beneficiary persons not post-shock poor. For an objective linear in shortfalls (P_1), the attainable share closely tracks the budget share reaching the poor. B1 and B3 tie on P_1 but differ on P_2 and on loss compensation.

6.3 Uncertainty (joint bootstrap U3, central scenario; paired differences vs R3)

Rule	ΔP_1 [95% CI]	ΔP_2 [95% CI]	Δ loss comp. [95% CI]
R1 universal	+0.253 [0.237, 0.269]	+0.217 [0.203, 0.231]	−0.148 [−0.161, −0.135]
R2 geographic	+0.191 [0.177, 0.206]	+0.160 [0.147, 0.173]	−0.105 [−0.117, −0.093]
R4 exposure	+0.138 [0.116, 0.157]	+0.122 [0.102, 0.139]	−0.069 [−0.087, −0.052]
R5 PMT + exposure	−0.0004 [−0.003, 0.002]	−0.0003 [−0.003, 0.003]	+0.0008 [−0.003, 0.003]
B1	*−0.240 [−0.258, −0.224]*	*−0.661 [−0.685, −0.638]*	*−0.294 [−0.314, −0.275]*
B3	*−0.235 [−0.253, −0.210]*	*−0.102 [−0.119, −0.083]*	*+0.547 [0.504, 0.575]*

- **Levels:** the SE of P_1 is about 0.44, and of P_2 about 0.25.
- **Best-rule probabilities (U3):** R5 is best in 62% of replicates on P_1, 63% on P_2 and 76% on loss compensation. R3 takes the rest; no other rule is ever best.
- **Other scenarios:** $\lambda = 0$ and $\lambda = 1$ give the same ordering.
- **Model refitting** adds ≤ 1% to the SEs for P_1 and P_2, and up to 33% for loss compensation.
- **Information error alone (U2)** has an SD of 0.003–0.008 points of P_1.

6.4 Sensitivity grid (972 scenarios)

- **Best implementable rule:** R5 in 86% (P_1), 97% (P_2) and 84% (loss compensation) of scenarios; R3 in the rest; no other rule ever best.
- **Ranking stability:** R2 beats R1, and R4 beats R2, in every scenario.
- **Shock size:** θ moves it by about 45%, λ by about 8%, seasonal normalisation by about 3% and ε by about 7%. None of these changes the ranking.

6.5 District diagnostics

- **Between-district R^2:** 0.079 for $\Delta y/c_0$, 0.095 for the post-shock poverty gap and 0.059 for the change in the gap (0.086, 0.108 and 0.065 with district × urban/rural).
- **Top-8 probability:** Mulanje 1.00, Phalombe 0.95 and Nsanje 0.93. No other district exceeds 0.65.
- **Precision:** every district estimate rests on at least 16 EAs and 256 households.

7. Validation and reproducibility

- **Synthetic-fixture tests (10 tests, 80 expectations; no microdata)** verify:
  - welfare-change algebra;
  - FGT measures;
  - equal budgets for every rule under arbitrary weights;
  - B1 attains the analytic maximum P_1 reduction $G/(zN)$ and the minimum P_2;
  - B3 attains the maximum loss compensation;
  - no own-EA leakage in cross-fitting;
  - Duan smearing ≥ 1;
  - coverage targeting;
  - refusal of unsafe output locations;
  - fail-safe Stage 1 checks;
  - whole-EA, within-stratum resampling.
- **Run discipline:** each run writes to a new timestamped directory with a manifest recording code file hashes, input hashes, parameters, seeds, R and package versions, test and gate status, and output hashes. Scripts cannot write to the validated tables.
- **Comparison with the frozen pre-audit tables:** every quantity whose definition was unchanged reproduces to machine precision. R5 moved by ≤ 0.006 after Duan smearing was added. The B1 − R3 confidence interval

narrowed after the budget was re-normalised per replicate: [0.22, 0.26], previously [0.14, 0.34].

8. Limitations

1. **Maize-specific price deflation (IFPRI).** IHS6 fieldwork coincided with an extreme maize price spike: IFPRI reports about K600/kg in June 2024 and about K1,700/kg by March 2025, while the general temporal deflator rose about 37%. Purchases are deflated with general CPI, so real maize spending in lean-season interviews, and therefore normal-year exposure, may be overstated. A maize-specific monthly deflator from IFPRI's market price series was not used: the external data were not obtained.
2. **Registry realism.** R3–R5 are trained and evaluated on the same survey round, with covariates measured in the same visit. A deployable registry model would be trained on older data and applied to separately collected records with staleness, misreporting and partial coverage. Malawi evidence (Gualavisi & Newhouse 2025) shows 2017 registry PMT scores ranking village welfare poorly, and Brown, Ravallion & van de Walle (2018) find PMTs exclude many of the poor. The PMT advantage reported here is therefore an upper bound. An IHS5→IHS6 transport test was designed but not run.
3. **District-only geography.** Geographic rules use 32 districts. Traditional Authority or village rules would need census microdata (IPUMS licence not applied for) or registry data. The weak district result should not be generalised to finer geographic targeting, which performs much better in Malawi when combined with a partial registry and geospatial data.
4. **Bootstrap limitations.**
   - Replicate weights are not re-calibrated to the population projections used in the IHS6 weights.
   - No finite-population correction is applied.
   - Cross-fitting fold assignment is fixed.
   - Scenario parameters are varied deterministically, not sampled.
   - Household measurement error (recall, seasonality, sales timing) is not part of the uncertainty.
   - Percentile intervals use 200 replicates.
5. **Measurement.** 7-day purchase recall annualised, with strong seasonality: the share buying is 33% in May and 71% in February. A third of sale records predate the fieldwork window and are assigned the window-start index.
6. **Scope.** Partial equilibrium only: no wage, supply-response, input-market or general-equilibrium effects, and no causal identification of price transmission. IHS5 and IHS6 comparability is unresolved, so no trends are claimed.

References: Araar & Verme (2012) SUBSIM, World Bank · Brown, Ravallion & van de Walle (2018) *JDE* 134 · Chiwaula et al. (2024) IFPRI MaSSP WP45 · Corral et al. (2022) *Guidelines to Small Area Estimation for Poverty Mapping*, World Bank · Deaton (1989) *Economic Journal* 99 · Fuje & Pullabhotla (2020) World Bank PRWP 9436 · Gualavisi & Newhouse (2025) *WBER* 39(2) · NSO (2023) *Malawi in Figures* · NSO (2026) *IHS6 Monetary Poverty Findings*. Disclosure: all reported statistics are aggregates over ≥ 16 EAs. No household, EA or coordinate information is reported.